

A Factor Decomposition Method for Separating Store Substitution and Embedding Placement Effects in Vector Database Migration

Junha Yoon · Ohhyeon Gwon · Junhaeng Lee · Hyeji Roh · Soowan Cho · Suhee Kim · SangSoon Lim

Sungkyul University · Chung-Ang University · Deep Insight Lab

2026 International Conference on PyGeek

Hongik University

Contents

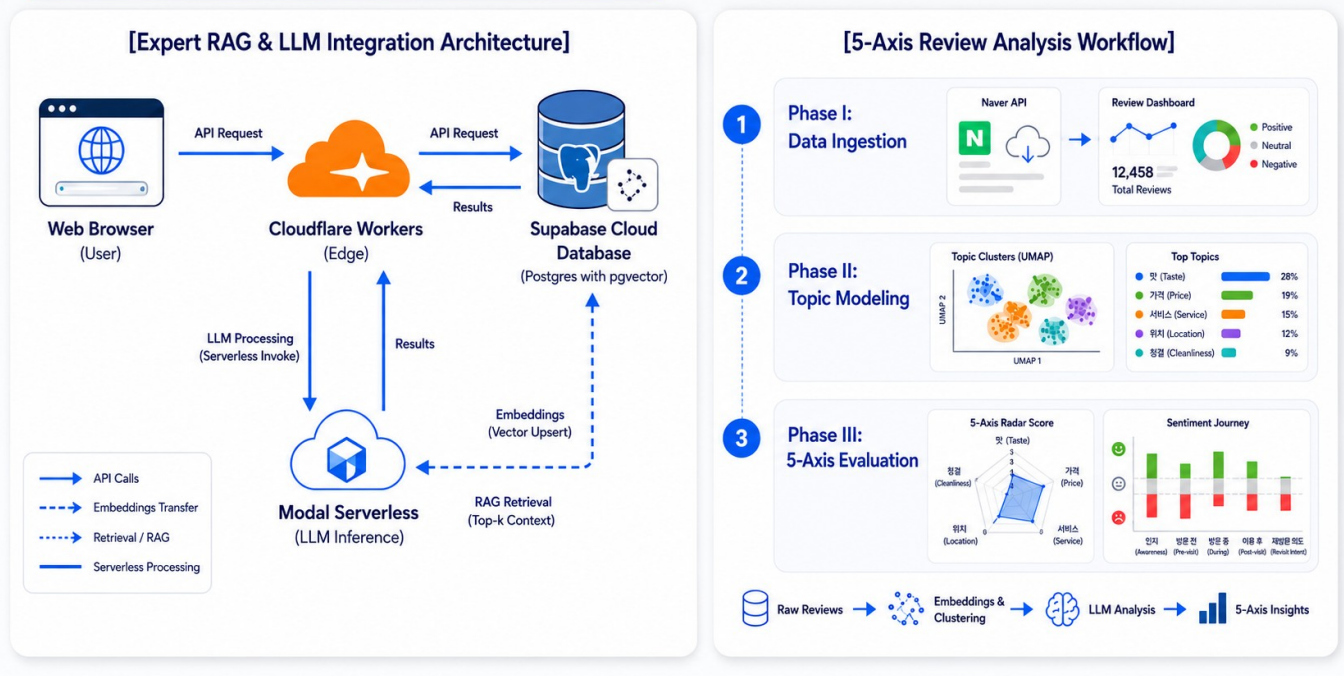
- 1 Introduction**
- 2 Proposed Method**
- 3 Performance Evaluation**
- 4 Experiment Result**
- 5 Conclusion**

Introduction

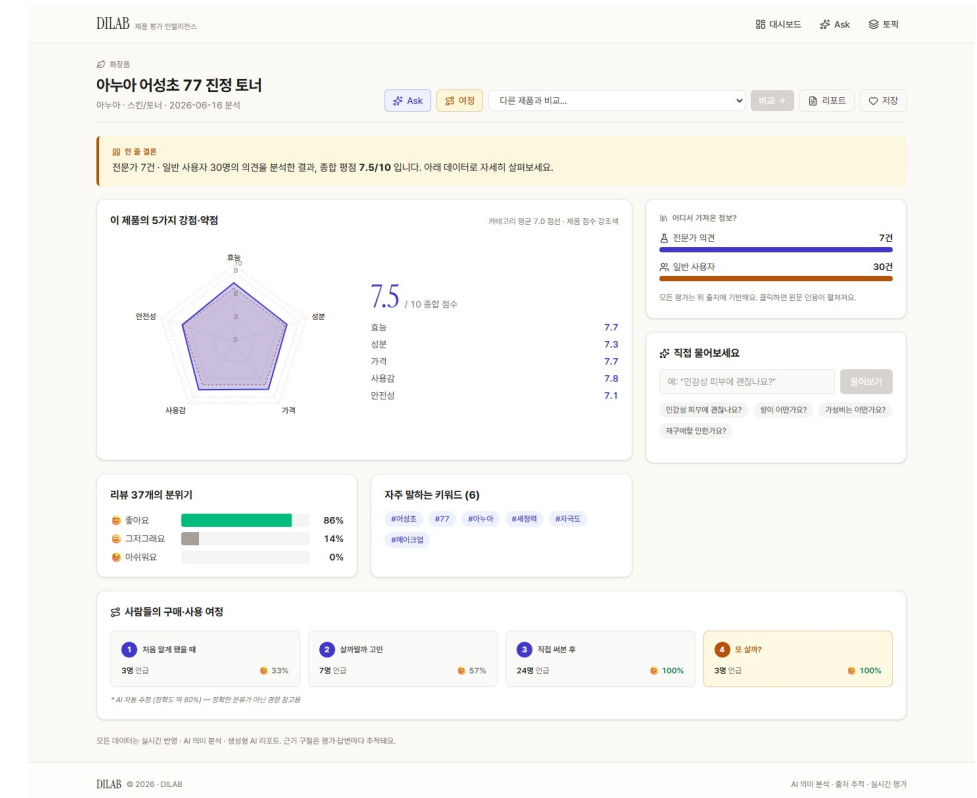
Background — DILAB, a product-review analysis service

- Consumer and expert reviews are embedded into a vector store and served as evidence-cited reports
- The prototype runs on pgvector; production runs on Oracle, so the vector layer must migrate

DILAB Core System & AI Pipeline



System architecture & analysis pipeline

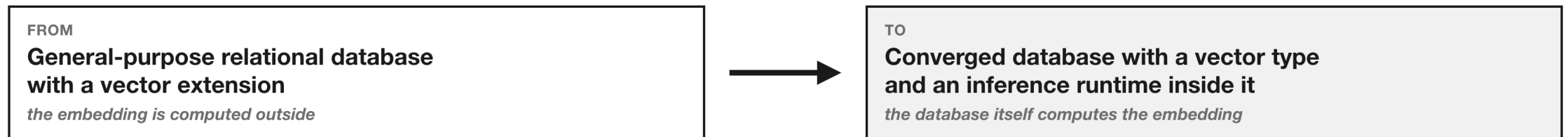
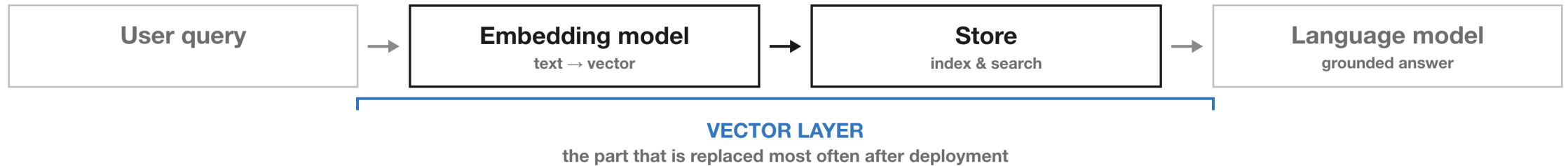


Product report dashboard

Introduction

The vector layer is the part that gets replaced after deployment

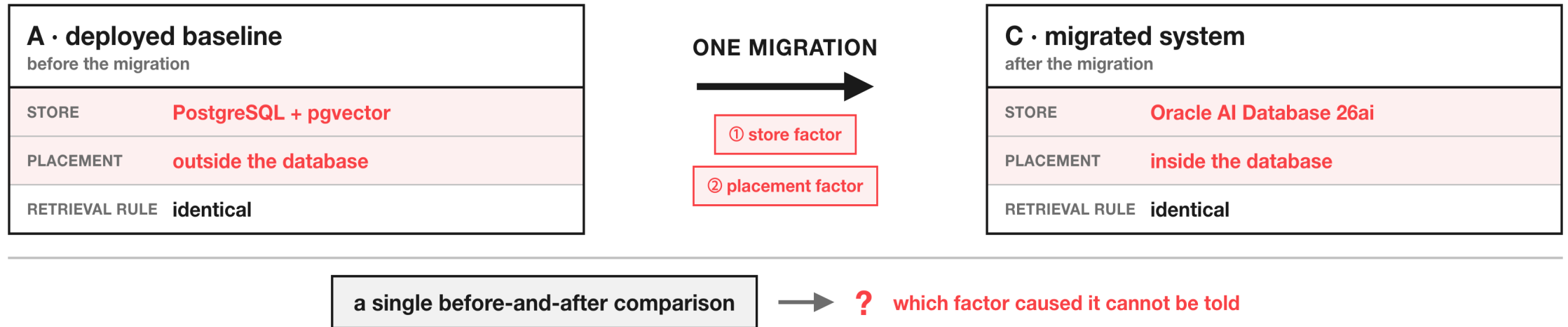
- A RAG system combines an embedding model, a store, and a language model
- A common migration moves it to a converged database that embeds internally



Introduction

One migration changes two things at the same time

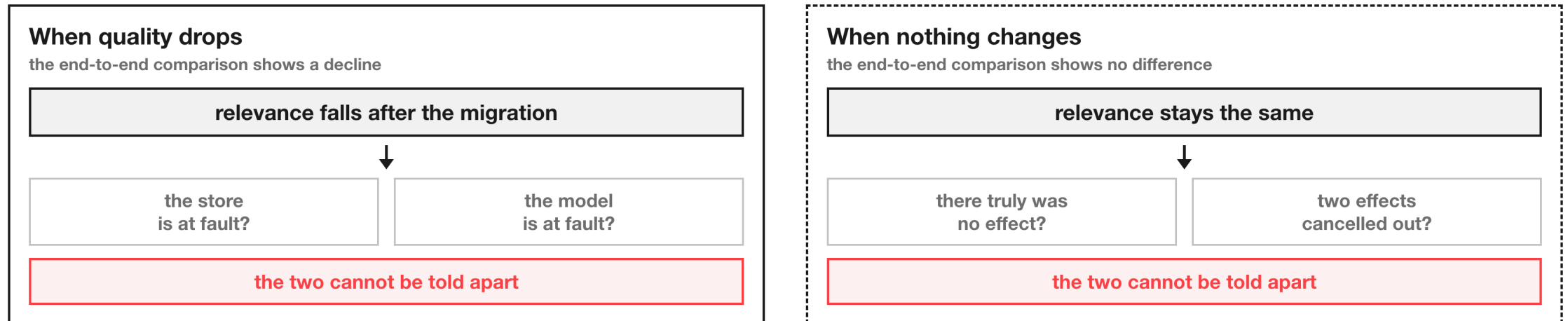
- The store and the embedding placement move together, not one at a time
- A governance-driven migration pulls the model inside, and the model size limit follows
- A single before-and-after comparison cannot attribute the change to either factor



Introduction

Misattributing the cause leads to the wrong remedy

- A decline cannot be assigned to the store or to the model
- No observed change may mean no effect, or two effects cancelling each other out

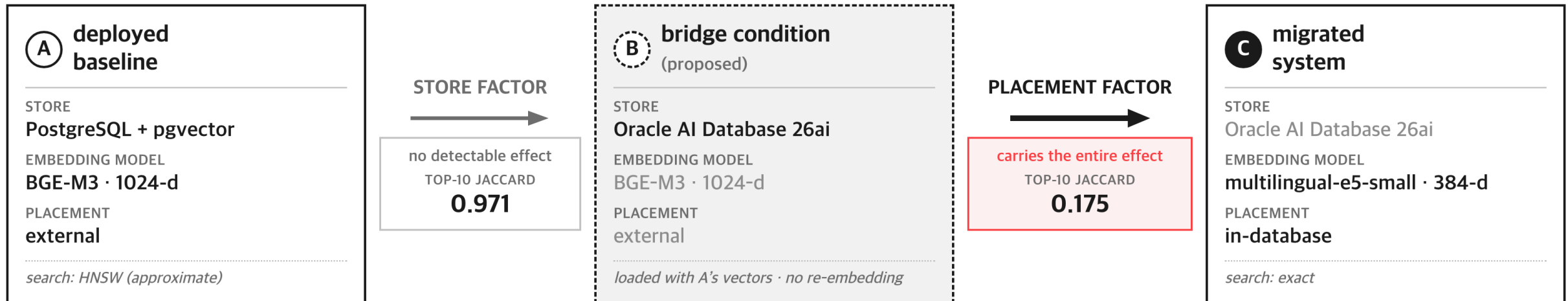


Blaming the store and reverting it does not bring the quality back
and the conclusion cannot be carried over to a different instance class or model size limit

Proposed Method

A bridge condition splits one comparison into two

- Load the target store with the vectors the source system has already produced
- A to B differs only in the store, B to C only in the placement
- No re-embedding is required, so the running service is left untouched



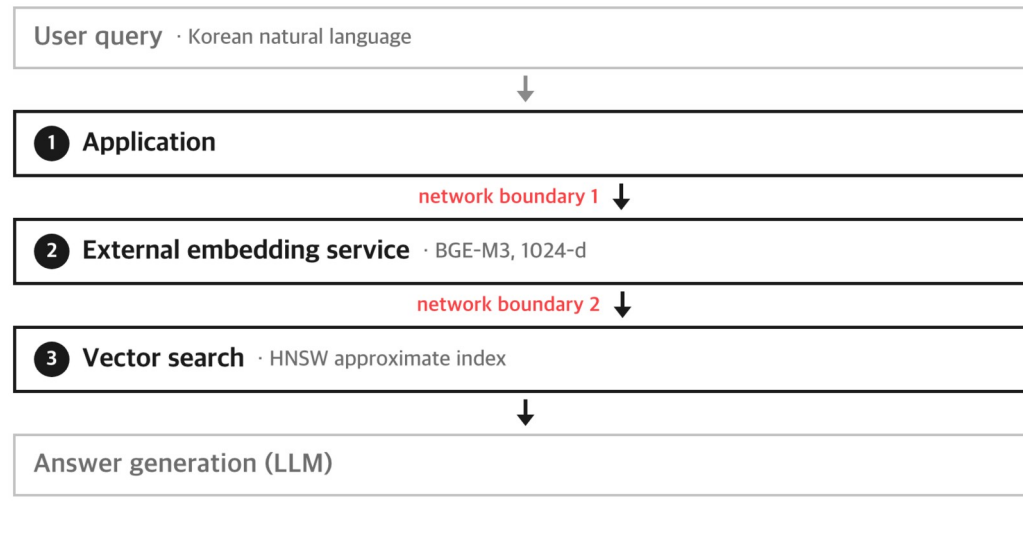
Proposed Method

Network boundaries drop from two to one and the external model call disappears

- The baseline crosses two network boundaries to reach an external embedding service
- After migration, embedding and search both run inside the database boundary
- The retrieval rule and its parameters were held identical across the three conditions

Deployed baseline

PostgreSQL + pgvector



Network boundaries

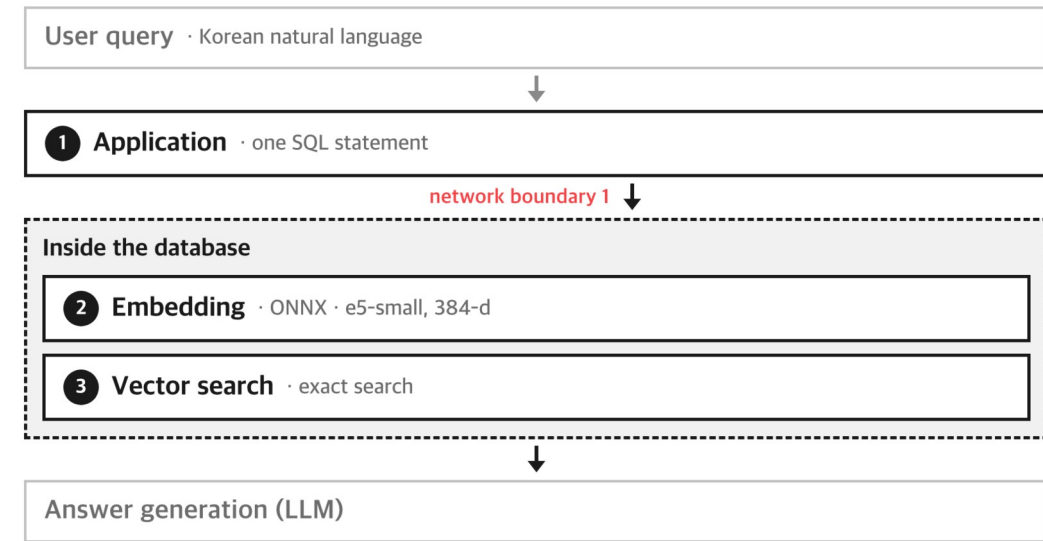
2

External model call

yes

Migrated system

Oracle AI Database 26ai



Network boundaries

1

External model call

no

Performance Evaluation

Three conditions measured over 1,347 chunks and 43 fixed queries

- Retrieval agreement, evidence relevance on a 0 to 2 scale, and search latency
- Confidence intervals from 10,000 bootstrap resamples on an entry-level instance

	A · baseline	B · bridge	C · migrated
Store	PostgreSQL + pgvector	Oracle 26ai	Oracle 26ai
Vector search	HNSW approximate	exact	exact
Embedding model	BGE-M3 · 1024-d	BGE-M3 · 1024-d	e5-small · 384-d
Placement	outside the database	outside the database	inside the database

→ STORE FACTOR → PLACEMENT FACTOR

■ changed by the store factor ■ changed by the placement factor □ carried over unchanged

Experiment Result

How the two metrics are read

- Jaccard measures how much the two top-10 sets overlap, on a 0 to 1 scale
- The p value is how well chance alone explains the observed difference
- Here both conditions returned the same list and the difference sits mid-distribution

Top-10 Jaccard · 0.971

Condition A — top 10



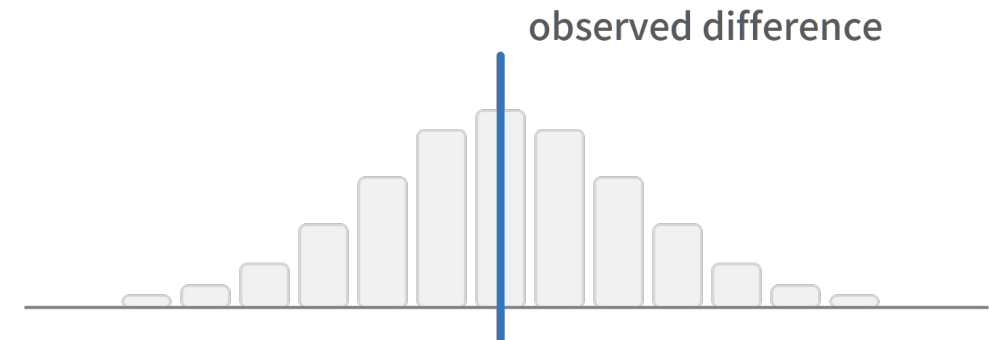
Condition B — top 10



Same list: $10 \text{ shared} \div 10 \text{ combined} = 1.00$

Identical top-10 set for 37 of the 43 queries

Paired difference · p = 0.523



Sits in the middle of the chance distribution

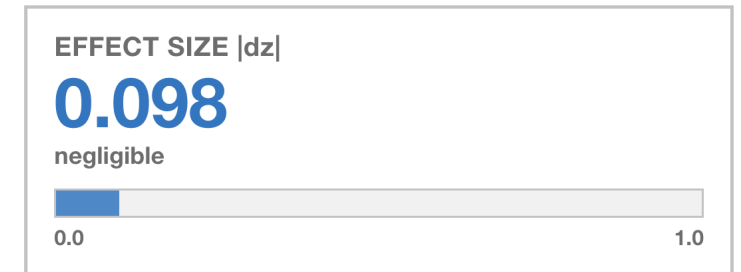
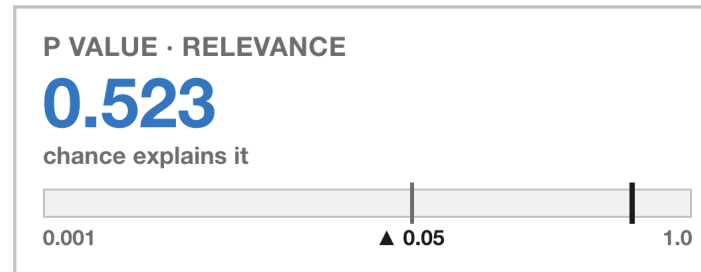
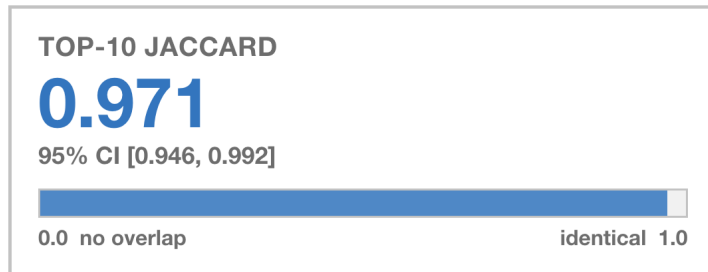
Chance explains it — no effect detected

Grey bars = the differences pure chance could produce when there is no real effect

Experiment Result

Store factor — no quality change was detected

- An identical top-10 set was returned for 37 of the 43 queries
- What the approximate index missed amounts to about 1.6 percent
- A Wilcoxon test on the queries remaining after ties agrees ($p = 0.908$)

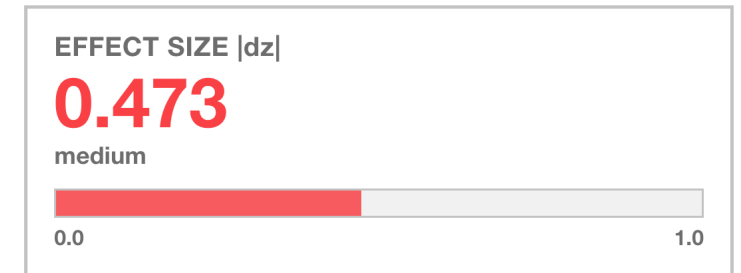
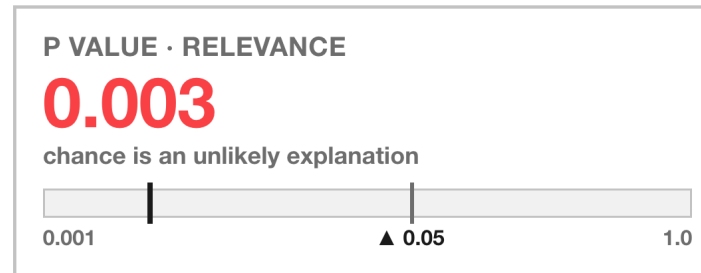
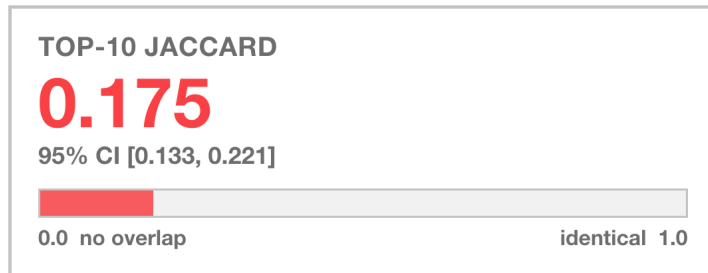


STORE FACTOR — no quality change was detected

Experiment Result

Placement factor — a significant, medium-sized effect

- The same store and the same search method returned mostly different evidence
- Since no store effect was detected, the change is attributable to the placement factor
- An end-to-end comparison alone would have blamed the store instead



PLACEMENT FACTOR — a significant, medium-sized effect

Experiment Result

The same two metrics applied to the placement factor

- Jaccard falls from 0.971 to 0.175 and the p value from 0.523 to 0.003
- Only 3 items are shared between the two top-10 lists
- The observed difference now sits far out in the tail of the chance distribution

Top-10 Jaccard · 0.175

Condition B — top 10



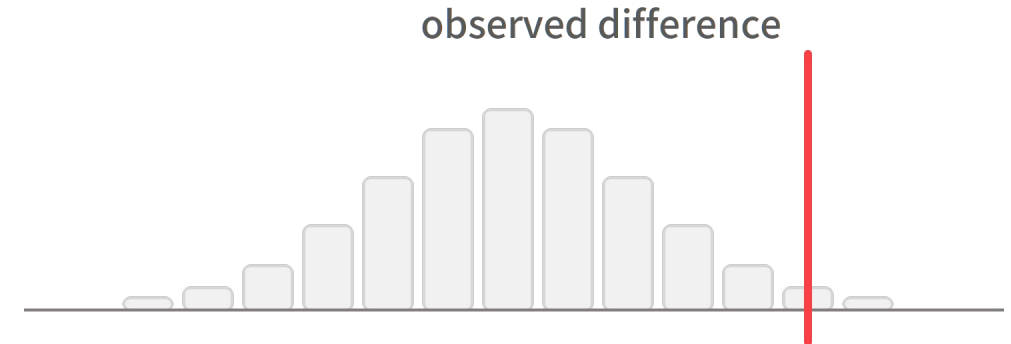
Condition C — top 10



3 shared ÷ 17 combined ≈ 0.18

Mostly different evidence for the same query

Paired difference · p = 0.003



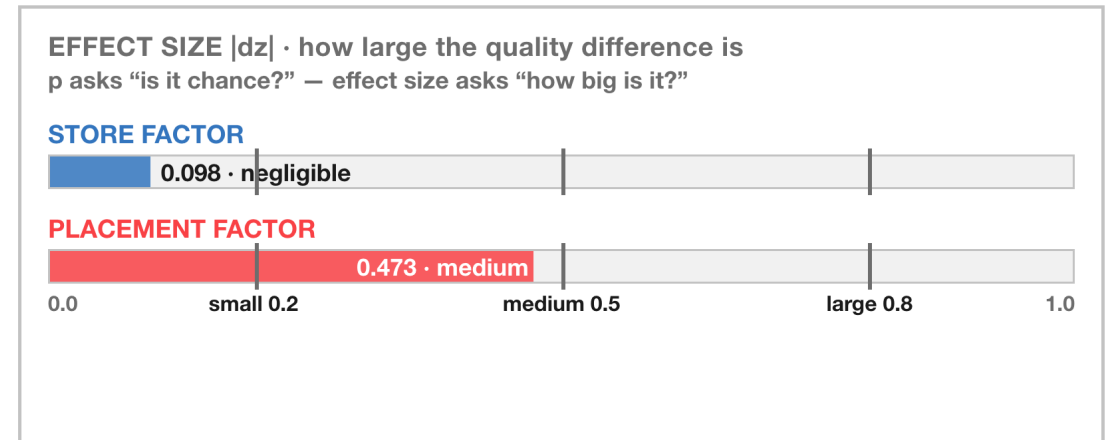
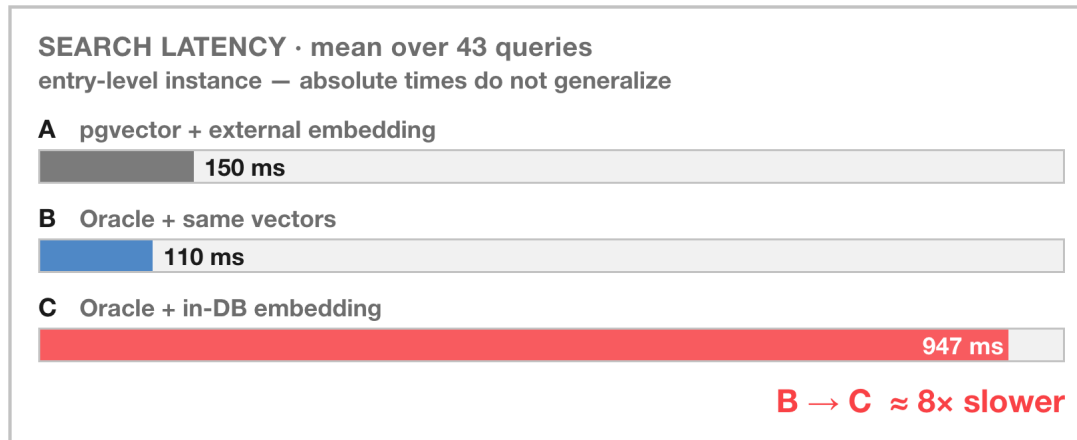
Sits far out in the tail of the distribution
Chance is an unlikely explanation

Grey bars = the differences pure chance could produce when there is no real effect

Experiment Result

Search latency and effect size point to the same factor

- Replacing the store left search a little faster — 150 ms down to 110 ms
- Moving the embedding inside made search about eight times slower — 110 ms to 947 ms
- Effect size states how large a difference is — negligible for the store, medium for the placement

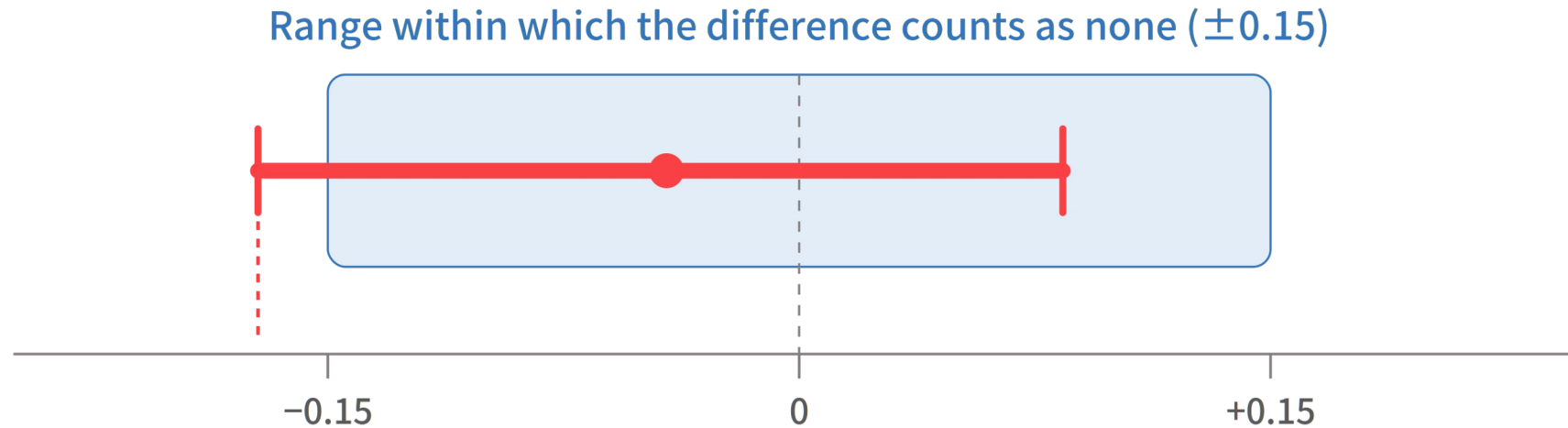


BOTH THE SLOWDOWN AND THE QUALITY DROP BELONG TO THE PLACEMENT FACTOR

Experiment Result

Claiming no difference requires an equivalence test of its own

- A non-significant p value is not evidence of equivalence
- TOST over a plus or minus 0.15 range gave $p = 0.052$, so the level was not met
- We report no effect detected with a bounded magnitude, not equivalence established



95% CI $[-0.172, 0.084]$ · point estimate -0.042

The left end falls outside the range → equivalence not established ($p = 0.052$)

Conclusion

Moving the placement is a trade in governance rather than in performance

- A single bridge condition splits an end-to-end comparison into two single-factor comparisons
- No store effect was detected; the entire observed change came from the placement

A practical decision rule

- If governance of the stored data is the only goal, replacing the store alone is enough
- If the query path must stay inside, confirm the model size limit before migrating

References

- [1] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.
- [2] Y. Gao et al., “Retrieval-augmented generation for large language models: A survey,” *arXiv preprint arXiv:2312.10997*, 2023.
- [3] Oracle, “Oracle AI Vector Search User’s Guide,” *Oracle AI Database 26ai documentation*, 2026.
- [4] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, and F. Wei, “Multilingual E5 text embeddings: A technical report,” *arXiv preprint arXiv:2402.05672*, 2024.
- [5] pgvector, “pgvector: Open-source vector similarity search for Postgres,” *GitHub repository*, 2026.
- [6] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 2318–2335, 2024.
- [7] Y. A. Malkov and D. A. Yashunin, “Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 4, pp. 824–836, 2020.
- [8] L. Zheng et al., “Judging LLM-as-a-judge with MT-Bench and Chatbot Arena,” *Advances in Neural Information Processing Systems*, vol. 36, *Datasets and Benchmarks Track*, 2023.
- [9] D. Lakens, “Equivalence tests: A practical primer for t tests, correlations, and meta-analyses,” *Social Psychological and Personality Science*, vol. 8, no. 4, pp. 355–362, 2017.

Thank You

Contact to me: wmsgk111400@sungkyul.ac.kr